QuickBite Delivery Intelligence

End-to-end data analytics project deck

Project focus: data cleaning, EDA, SQL insights, Folium map, Plotly dashboard, and classification for late delivery prediction.

Dataset	Cities	Model	Accuracy
3,200 orders	6 Indian cities	Random Forest	88.5%

Prepared as a complete presentation package for academic submission.

Executive Summary

Volume	Revenue	Best city	Risk zone
3,200 orders	INR 9.28L	Mumbai	Surat

Traffic and weather are the strongest operational drivers of delay.
Mumbai leads in revenue and has the lowest late-delivery rate.
Surat and Pune need tighter SLA control and better routing.
The classifier reaches 0.885 accuracy and 0.962 ROC-AUC.

Key deliverables

Cleaned dataset with geospatial fields and delivery KPIs
EDA visuals for demand, value, rating, and lateness
SQL-based business questions and answer tables
Folium map and Plotly dashboard preview
Late-delivery classification with feature importance

Introduction

Business problem

Operations teams need to know where, when, and why delivery delays happen.

Objective

Turn raw order data into decisions that improve SLA, routing, and customer satisfaction.

Track demand, revenue, and lateness across cities and time windows.
Use SQL to answer repeatable business questions fast.
Visualize city hotspots and traffic pressure on a map.
Build a classification model to predict late orders early.

Success metrics

Faster decisions, fewer late orders, and a usable analytics workflow.

Data Collection and Wrangling Methodology

Collect

Synthetic order records were generated with city, time, location, and service fields.

Validate

Missing values, duplicates, and impossible values were checked and removed.

Transform

Date fields, ratings, lateness label, and categorical encodings were standardized.

Store

Final data was saved as CSV and loaded into SQLite for SQL analysis.

Cleaning rules applied

Clipped negative values and outliers in price, distance, and delivery time.
Derived order_hour, weekday, weekend, and late_delivery features.
Unified city names and categorical labels for analysis and modeling.
Created lat/lon coordinates for map-based visualization.

EDA and Interactive Visual Analytics Methodology

1. Question
What drives delay and revenue?

2. Segment
By city, traffic, weather, and time

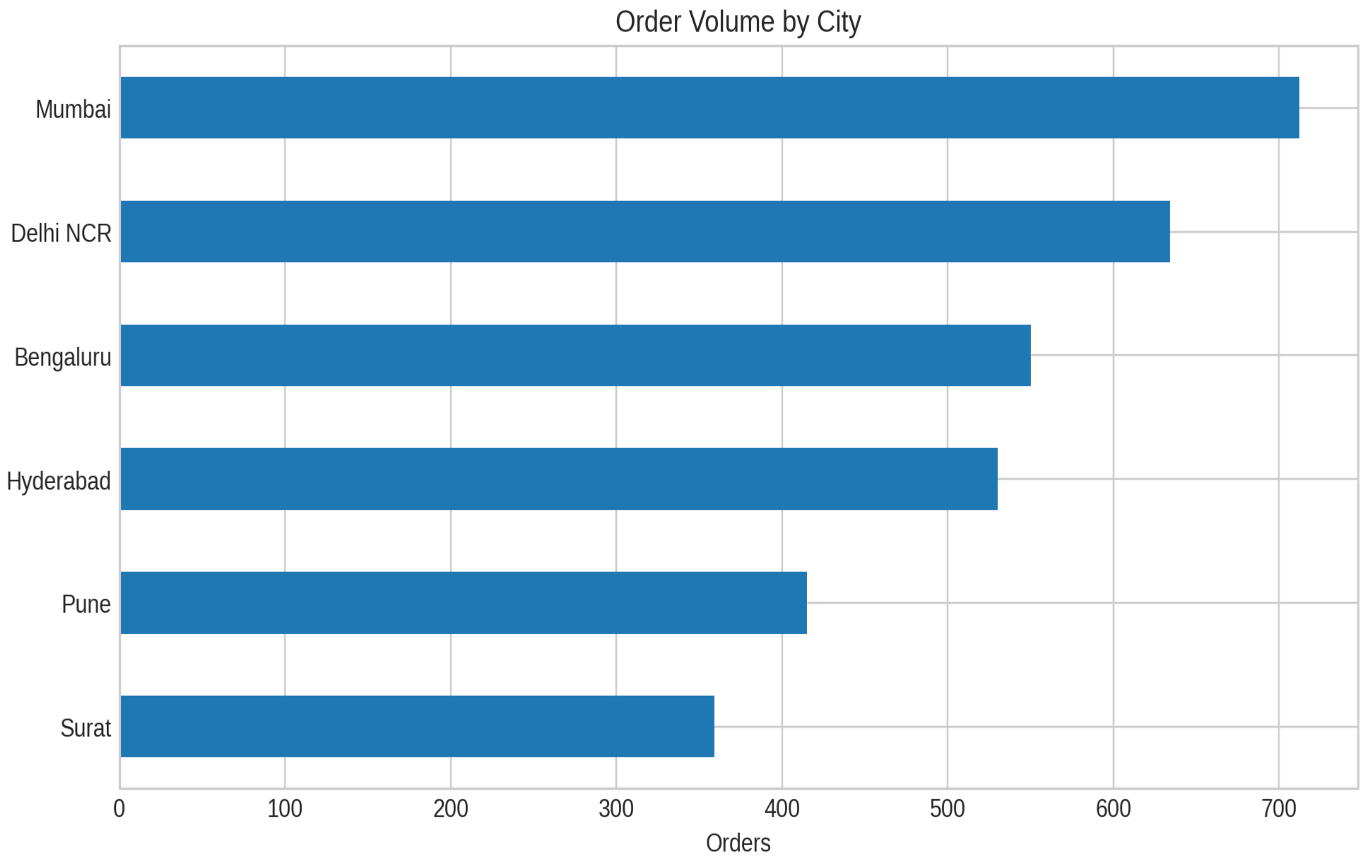
3. Visualize
Bars, lines, scatter, heatmaps

4. Act
Find fixes for SLA

EDA was used to spot patterns before modeling.
Interactive views were planned for city and zone-level filtering.
Charts were designed to answer one question per slide.
The output is easy to present and easy to extend.

Tools used
Pandas, Matplotlib, Seaborn, SQLite, Folium, Plotly

EDA Results 1

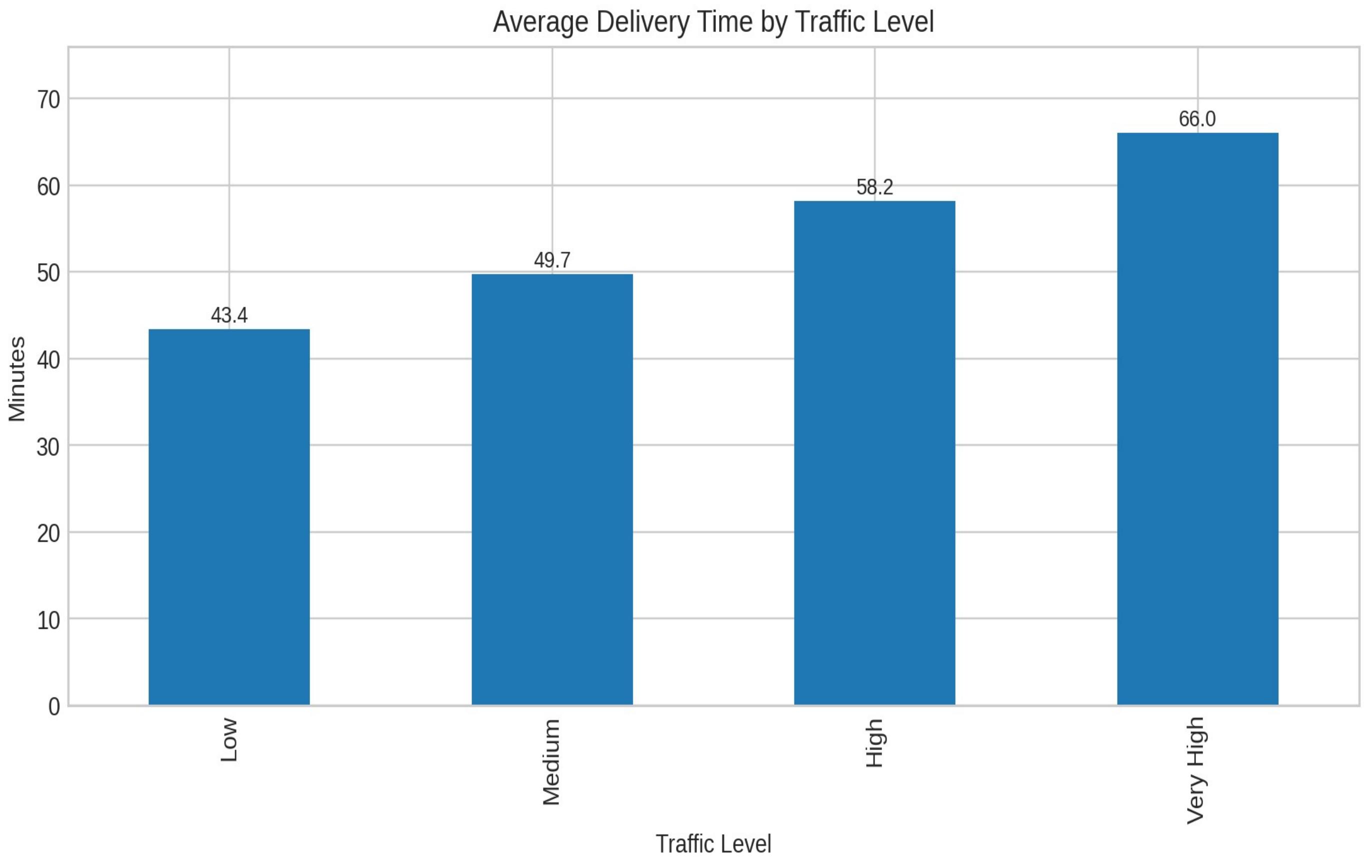


Top city
Mumbai

Lowest volume
Surat

Demand is concentrated in Mumbai, Delhi NCR, and Bengaluru.
Smaller cities can be targeted with local promos.

EDA Results 2

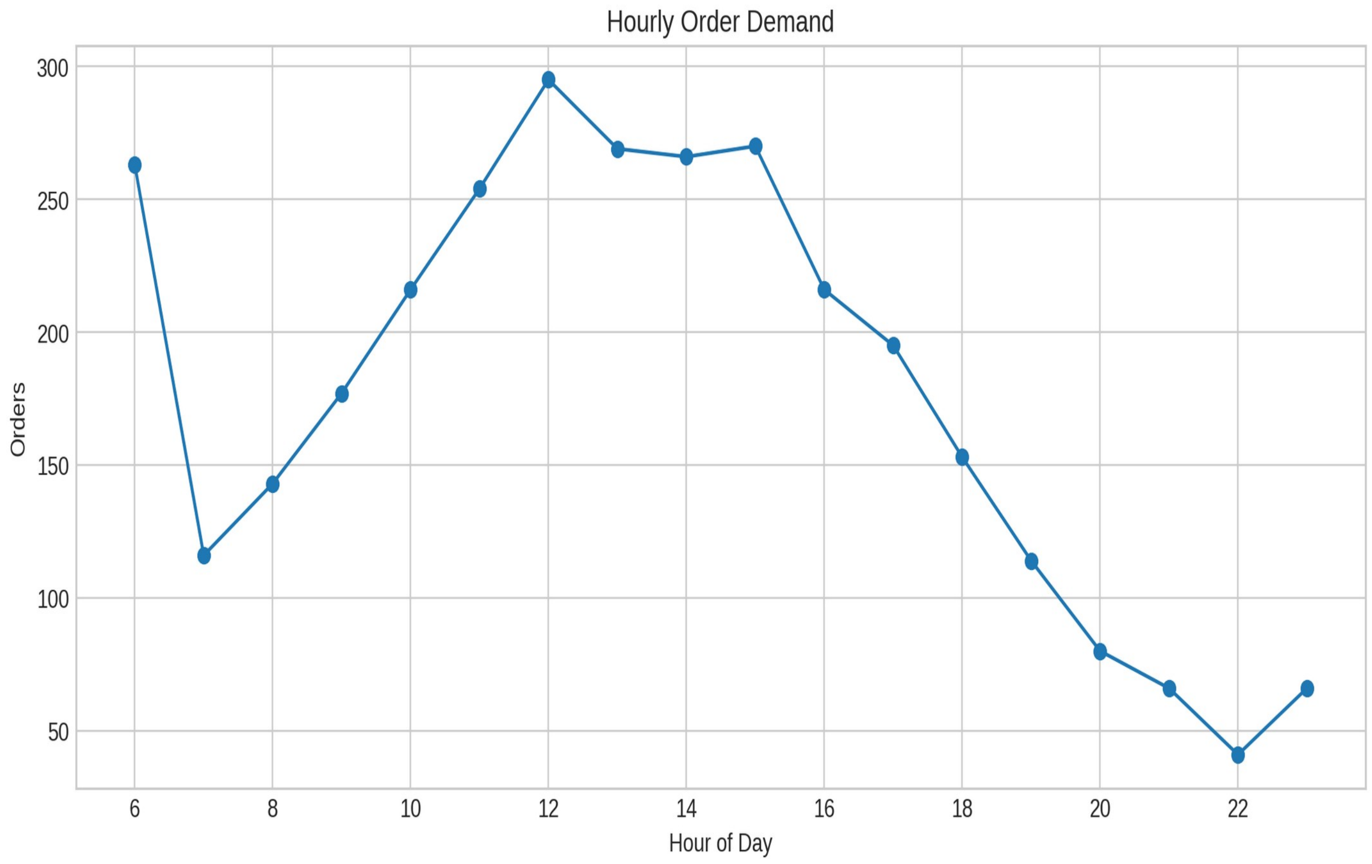


Low traffic averages about 43.4 minutes.
Very high traffic crosses 66 minutes.
Traffic control is the fastest lever for improvement.

Operational takeaway

Traffic-aware routing

EDA Results 3

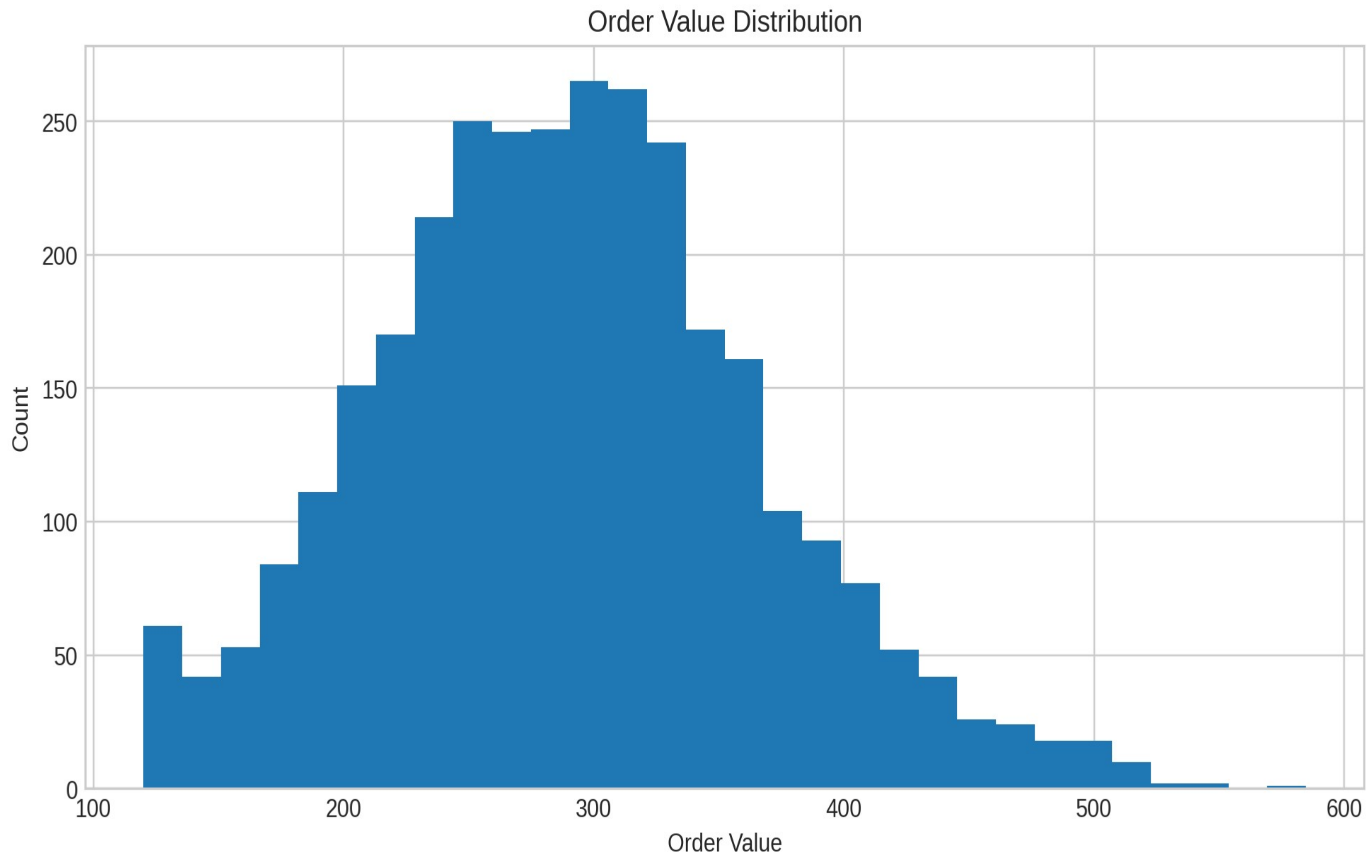


Orders peak around lunch and dinner hours.
Late-night demand is low and more variable.
Driver staffing should be shifted to the 12-14 and 19-21 windows.

Peak hours

12-14 and 19-21

EDA Results 4



Most baskets cluster around mid-range ticket sizes.
A small premium tail lifts revenue but also prep time.
Cross-sell bundles can raise average order value.

Revenue insight

Mid-range dominates

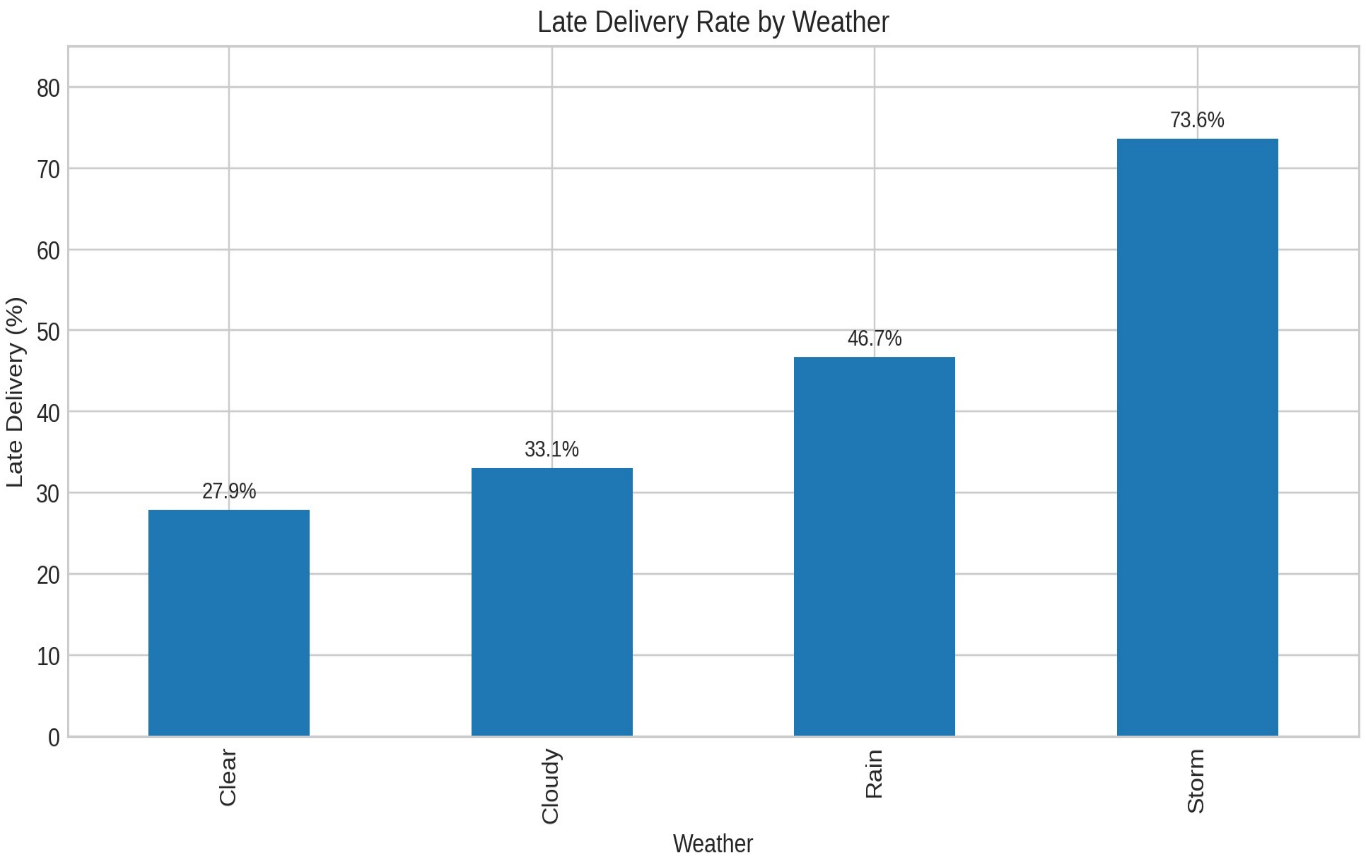
EDA Results 5

The scatter trends downward as delivery time grows.
Slow deliveries hurt ratings even when food quality is stable.
Delay control is also a customer experience strategy.

Risk signal

Long waits = lower ratings

EDA Results 6

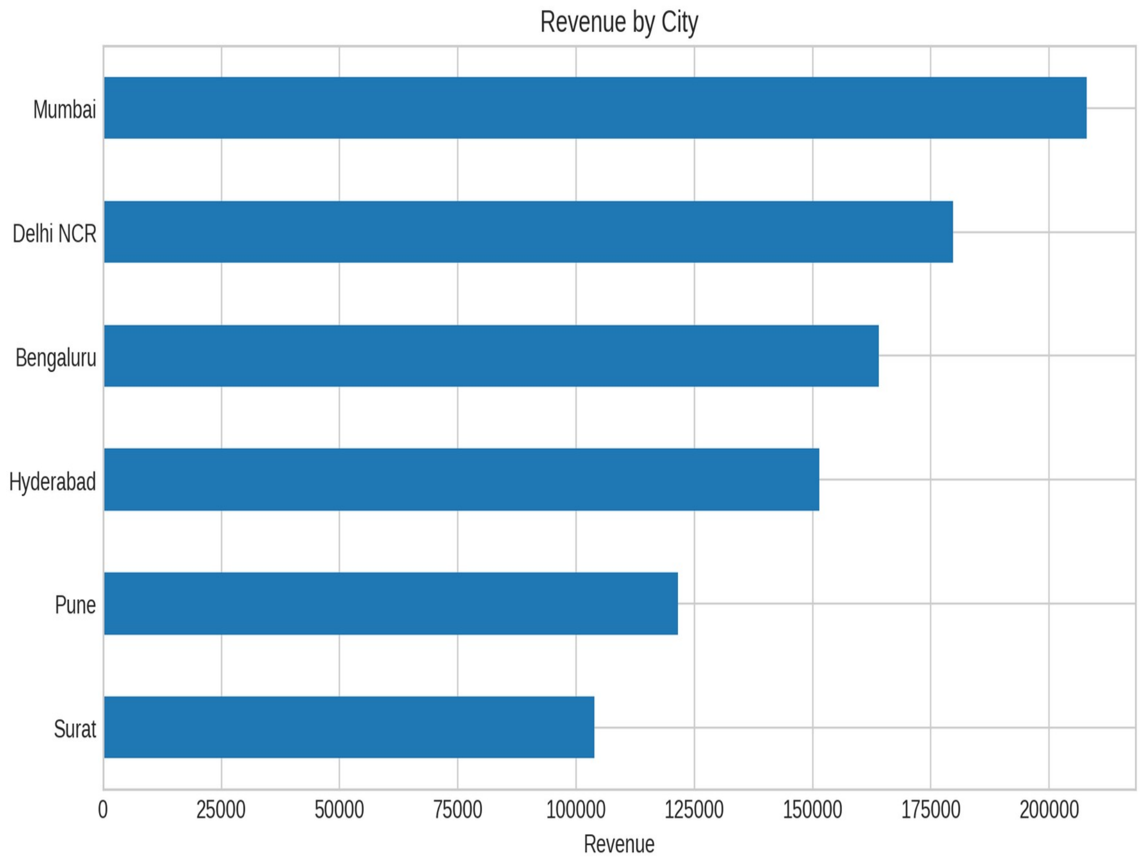


Storms are the worst condition for SLA breach.
Rain also raises lateness, but less severely.
Weather alerts should trigger routing buffers.

Best condition

Clear weather

EDA with SQL Results 1

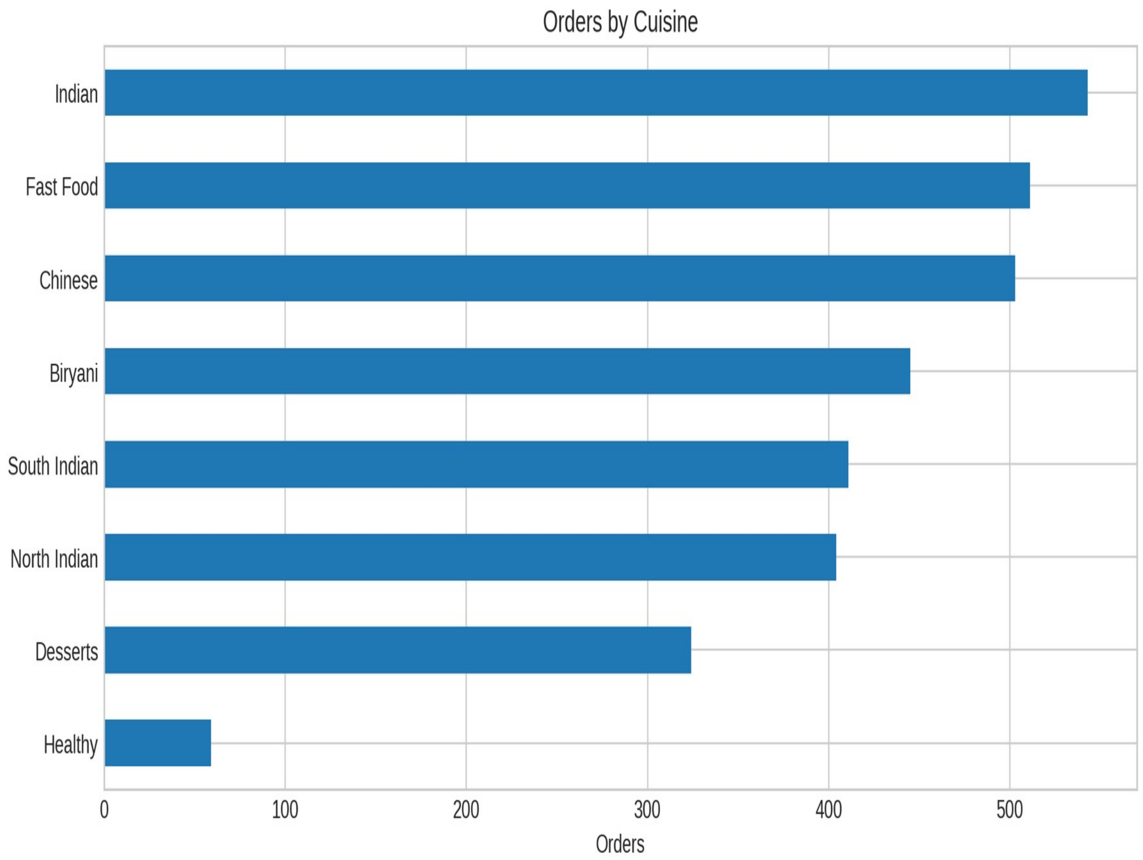


Top Cities by Revenue

	orders	revenue
Mumbai	712	INR 207,885
Delhi NCR	634	INR 179,647
Bengaluru	550	INR 163,938
Hyderabad	530	INR 151,355
Pune	415	INR 121,515
Surat	359	INR 103,897

Mumbai produces the highest revenue.
Surat contributes the least revenue and also needs demand growth.

EDA with SQL Results 2



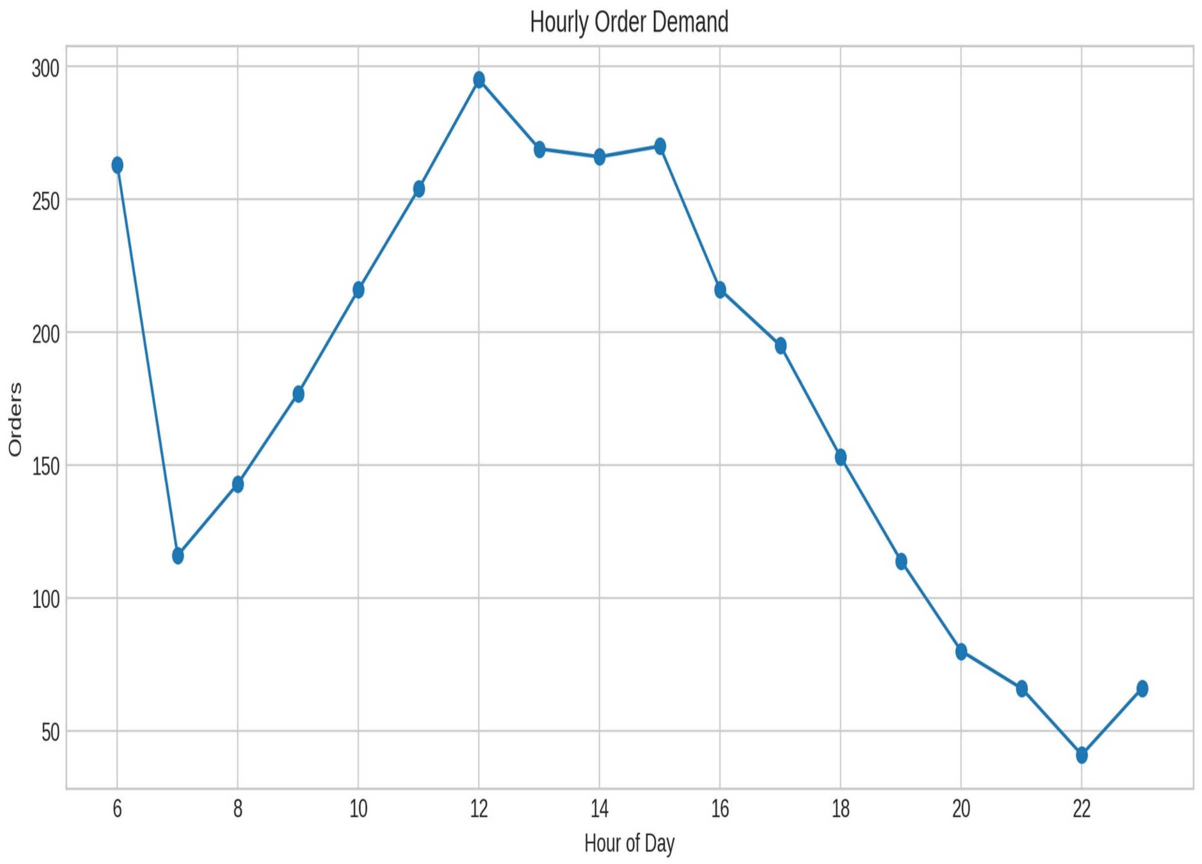
Top Cuisines

	orders	revenue	avg_rating
Indian	543	INR 170,577	3.77
Chinese	503	INR 147,582	3.74
Biryani	445	INR 145,502	3.81
Fast Food	511	INR 131,761	3.72
North Indian	404	INR 129,048	3.73
South Indian	411	INR 113,828	3.78

Indian and Chinese cuisines drive the biggest volume.
Biryani gives strong revenue with solid ratings.

EDA with SQL Results 3

Hourly peaks and operational timing



Top order hours

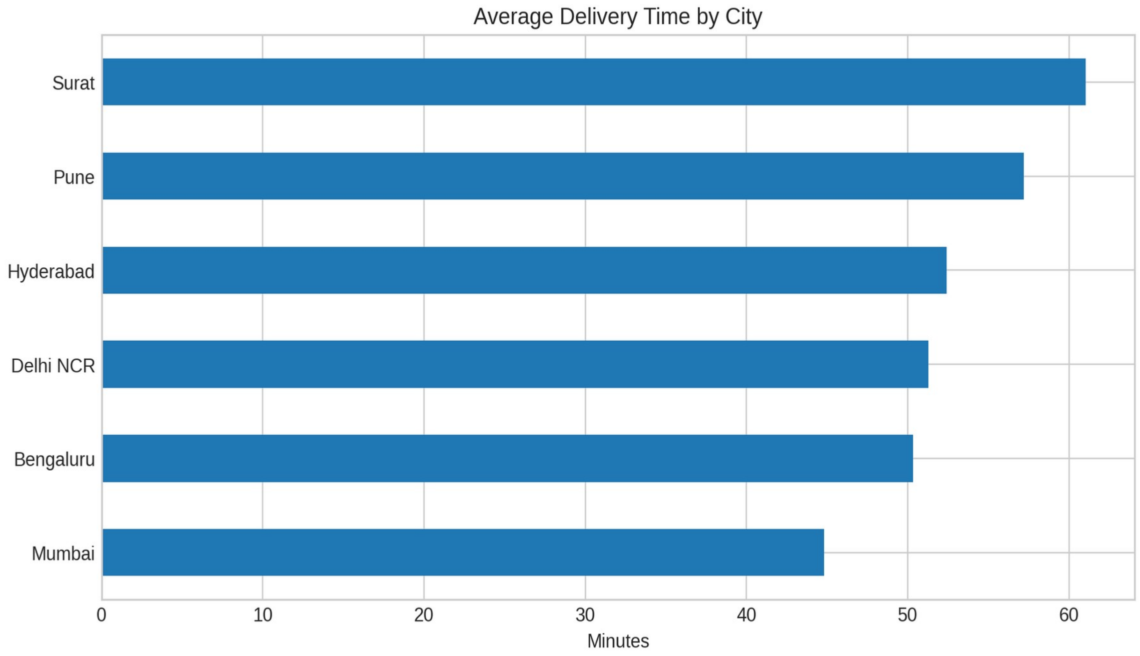
12-14 and 19-21

Lunch and dinner are the two demand spikes.
Mid-morning and late-night windows are much lighter.
Staffing should scale around these peaks.

EDA with SQL Results 4

Traffic Impact on SLA

	orders	avg_delivery	late_rate
Low	986	43.4	13.9%
Medium	1,091	49.7	29.5%
High	720	58.2	52.5%
Very High	403	66.0	73.4%



Average delivery time jumps sharply when traffic moves from Medium to Very High.
This is the clearest sign that routing and batching need attention.

EDA with SQL Results 5

Late Delivery Rate by City and Weather

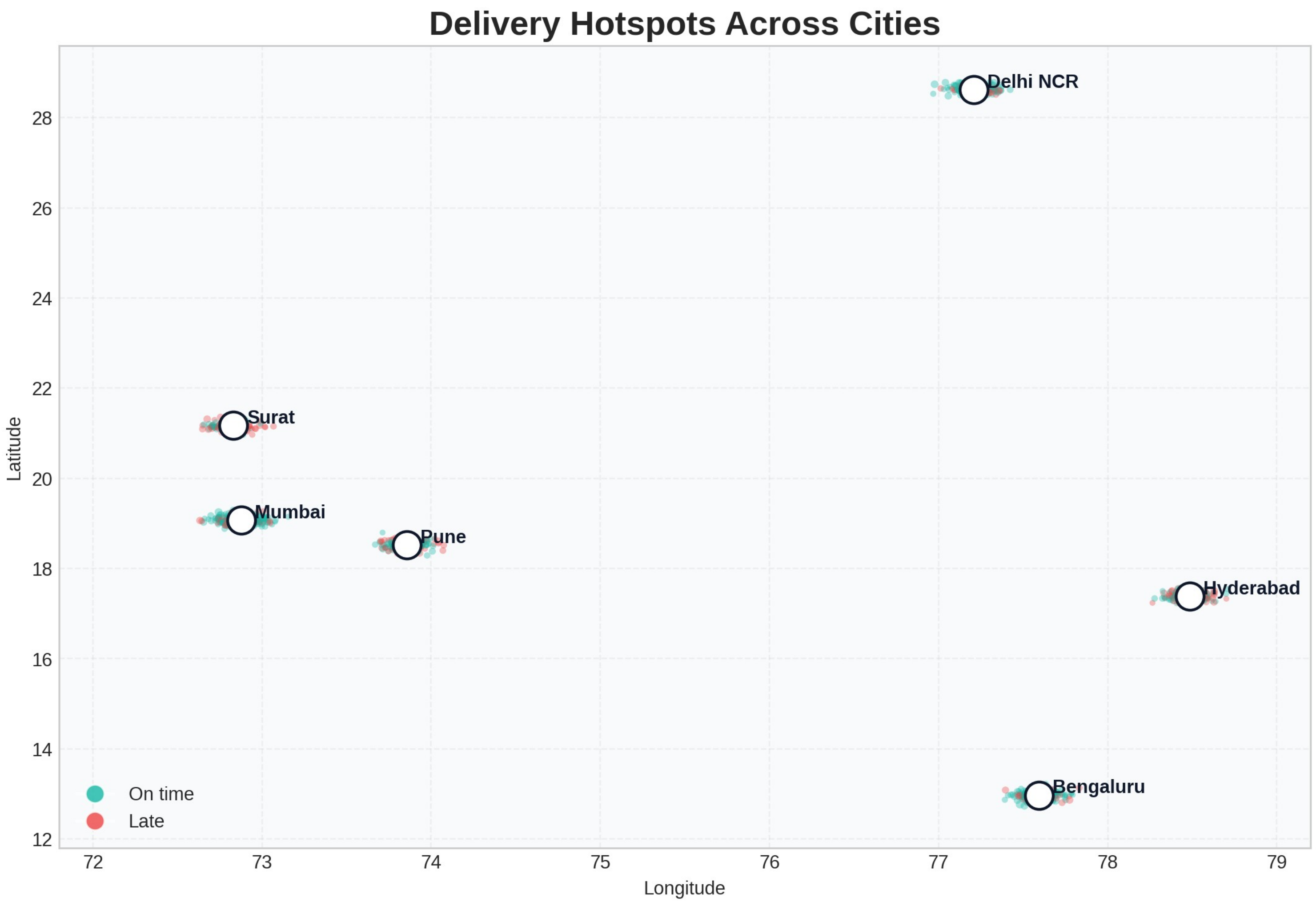
	Clear	Cloudy	Rain	Storm
Mumbai	11.7	9.0	30.3	58.7
Delhi NCR	23.7	32.4	47.7	76.1
Bengaluru	22.9	29.2	39.2	72.2
Hyderabad	30.8	38.1	46.5	70.0
Pune	44.5	56.3	58.3	88.2
Surat	51.2	54.1	74.6	92.9

Surat and Pune show the sharpest weather-linked lateness.
Mumbai is comparatively resilient in every weather class.
This matrix helps set city-specific buffer rules.

Best practice

Use city buffers

Interactive Map with Folium Results



Hotspots cluster around major city centers. Late orders are denser in longer-route zones. The interactive Folium HTML is included in the project folder.

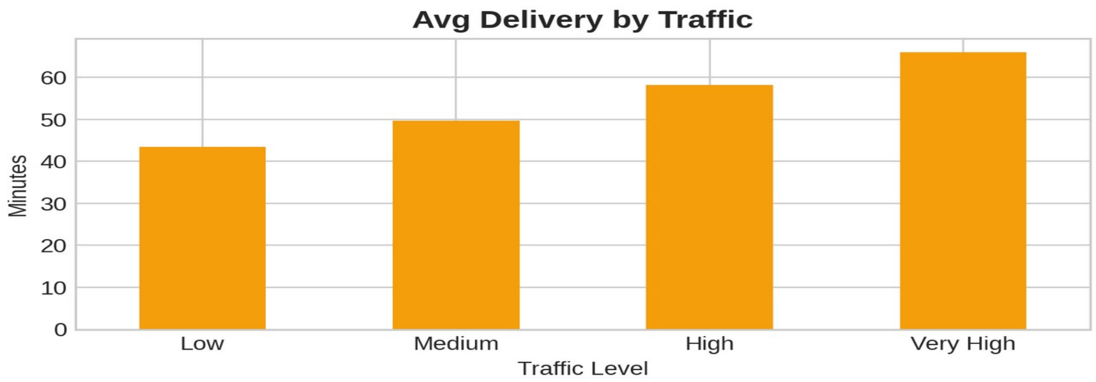
Map insight

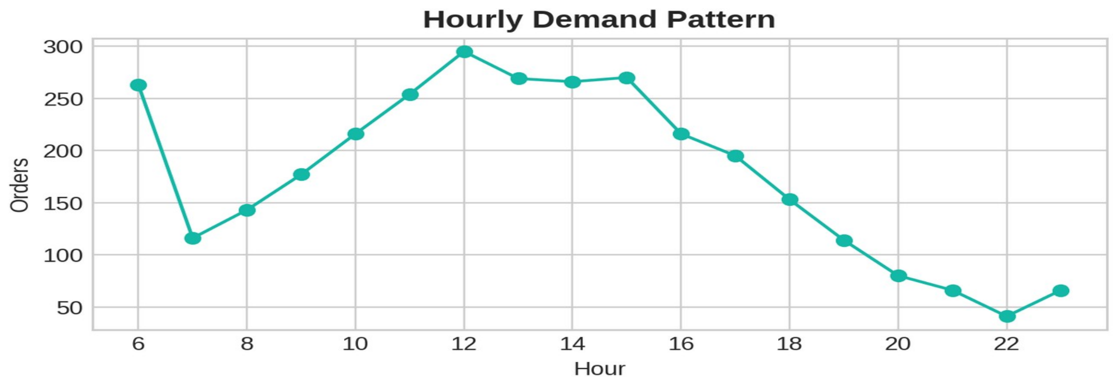
City hotspots are visible

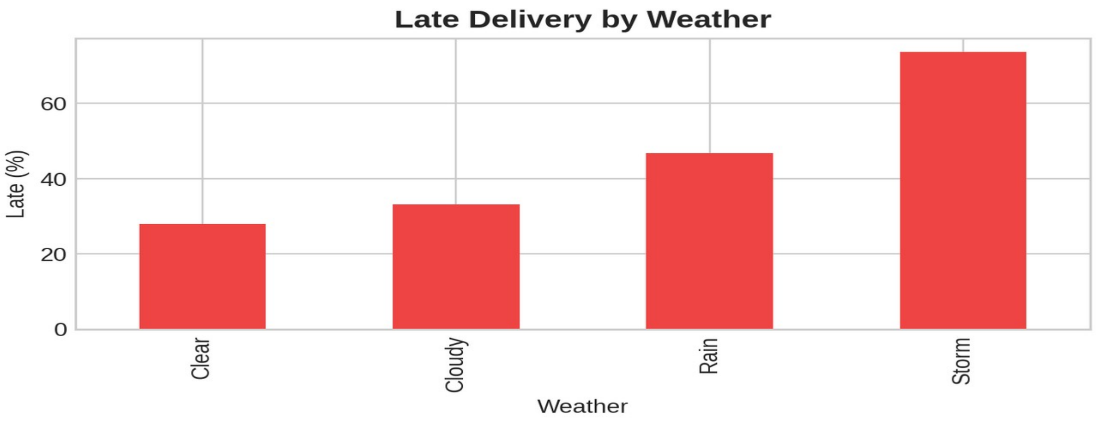
Plotly Dashboard Results

Operational snapshot

3,200	INR 9.28L	51.7 min	35.4%
Total Orders	Revenue	Avg Delivery	Late Rate







Predictive Analysis Methodology

Target
late_delivery

Model
Random Forest

Split
78 / 22

Encoding
One-hot

Features

City, restaurant type, cuisine, channel, payment, traffic, weather
Order hour, distance, prep time, order value, weekend flag

Why this model works

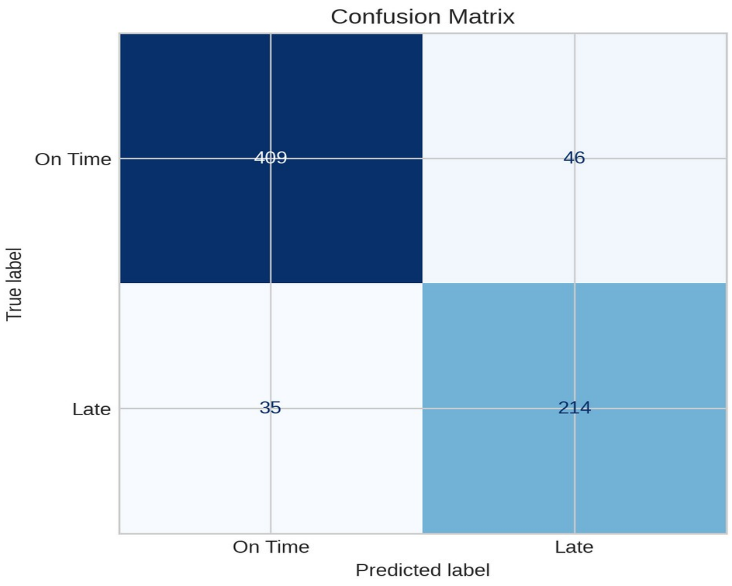
It handles mixed data types well and captures non-linear delay patterns.

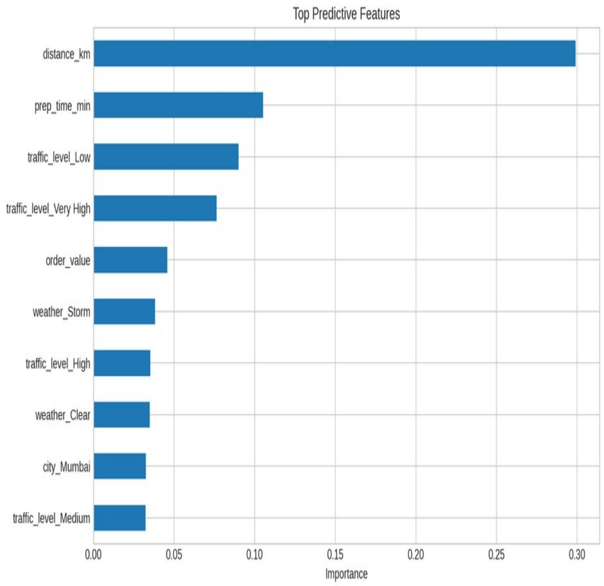
Predictive Classification Results

How well the model works

Model Performance

	Value
Accuracy	0.885
Precision	0.823
Recall	0.859
F1 Score	0.841
ROC-AUC	0.962





Distance and prep time are the dominant predictors.
Traffic and storm conditions push the model toward late delivery.
The model is strong enough to support early warning flags.

Creativity and Innovative Insights

1. SLA scorecards

Show city-level lateness daily so managers can react fast.

2. Weather triggers

Add automatic buffers when rain or storms appear.

3. Zone balancing

Shift drivers to hotspots before the late-order spike.

Build a simple SLA alert rule: if traffic is High and weather is Rain, escalate the order.
Prioritize late-risk cities first, not just the biggest cities.
Use the map and dashboard together to move from insight to action.

Conclusion

What the project shows

The workflow connects data cleaning, EDA, SQL, mapping, dashboarding, and classification into one story.

Operations should focus on traffic, weather, and route distance first.
Mumbai is the most stable city; Surat is the most vulnerable.
The predictive model can be used as an early warning layer.

Next step: replace the synthetic dataset with your real project data and keep the same presentation structure.